

# Unit 4 Assignment: Getting Started with Vertex AI Gemini API and Python SDK

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# Unit 4 Assignment Answer Sheet

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## Part A: Practice with Generative AI Tool

Screenshot: Text Prompt and Result

*“Explain how artificial intelligence can improve early detection of chronic diseases such as diabetes or heart disease. Include at least two specific examples.”*

### How AI Improves Early Detection of Chronic Diseases

Chronic diseases like diabetes and heart disease are among the leading causes of death worldwide, yet many cases go undetected until symptoms become severe. AI is transforming early detection by finding patterns in data that humans might miss — often years before a diagnosis would traditionally be made.

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### Why AI Is Well-Suited for This Problem

Early disease detection depends on recognizing subtle, complex signals across large amounts of patient data — lab results, imaging, genetics, wearable sensor readings, and medical history. Traditional screening relies on fixed thresholds (e.g., a specific blood sugar level), but disease risk is rarely that simple. AI excels at:

- **Pattern recognition** across many variables simultaneously
  - **Continuous monitoring** rather than point-in-time snapshots
  - **Personalization**, adjusting predictions to an individual's unique profile
  - **Scale**, screening millions of records faster than any clinical team
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### Example 1: Predicting Type 2 Diabetes Before Onset

Google and DeepMind have developed models that analyze retinal scans — photographs of the back of the eye — to detect early signs of diabetic retinopathy, but also to predict broader cardiovascular risk. More directly, AI systems trained on electronic health records (EHRs) can identify patients at high risk of developing Type 2 diabetes **2–5 years before** a clinical diagnosis.

A notable example is the work done with the **UK Biobank dataset**, where machine learning models analyzed blood biomarkers, lifestyle data, and genetic information to stratify individuals by diabetes risk far more accurately than the standard BMI-plus-fasting-glucose approach. This allows clinicians to intervene early with lifestyle modifications before the disease takes hold.

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**Example 2: Detecting Heart Disease from ECGs and Imaging**

The **Mayo Clinic** developed an AI algorithm trained on electrocardiograms (ECGs) that can detect **asymptomatic left ventricular dysfunction** — a precursor to heart failure — in patients who would otherwise show no symptoms. In clinical trials, this AI identified the condition with high accuracy from a standard 10-second ECG, something cardiologists alone could not reliably do.

Similarly, AI-powered analysis of cardiac MRI and CT scans can detect subtle structural changes in the heart (such as arterial plaque buildup) years before a heart attack risk becomes clinically apparent, enabling preventive treatment far earlier.

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**Broader Impact**

| Area      | AI's Role                                               |
|-----------|---------------------------------------------------------|
| Screening | Flags high-risk patients from routine data              |
| Imaging   | Reads scans with radiologist-level accuracy             |
| Wearables | Monitors heart rhythm, glucose trends in real time      |
| Genomics  | Identifies genetic predispositions early                |
| Access    | Extends specialist-level screening to underserved areas |

**Key Takeaway**

AI doesn't replace clinicians — it acts as a highly sensitive **early warning system**, surfacing risk signals that would otherwise be invisible until a disease progresses. For chronic conditions where early intervention dramatically improves outcomes (and reduces costs), this capability is genuinely life-saving. The ongoing challenge lies in ensuring these systems are trained on diverse, representative data to work equitably across all populations.

Screenshot: System Instructions and Result

***“You are a clinical decision support assistant. Your responses should be clear, concise, and focused on patient safety. Avoid speculation and only provide medically grounded explanations.”***

***“Explain how AI can assist doctors in prioritizing patients in an emergency department.”***

## **AI in Emergency Department Triage**

As a clinical decision support assistant, here's a medically grounded overview of how AI can help prioritize patients in the ED.

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### **The Core Problem AI Addresses**

Emergency departments face **high patient volumes, time pressure, and variable severity** — making triage one of the most consequential and error-prone steps in care. Traditional triage (e.g., ESI scoring) depends heavily on clinician experience and can miss subtle deterioration signals.

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### **How AI Assists**

#### **1. Rapid Risk Stratification**

AI models trained on large ED datasets can analyze **vital signs, chief complaint, age, comorbidities, and lab values** simultaneously to generate a risk score — faster and more consistently than manual assessment alone. This helps flag high-acuity patients who may not appear critically ill on first presentation.

#### **2. Early Warning of Deterioration**

AI can continuously monitor patients **already in the waiting room or under observation**, alerting staff when parameters trend toward sepsis, cardiac events, or respiratory failure — before a clinical threshold is formally breached.

#### **3. Sepsis and Stroke Detection**

Validated tools like **Epic's Sepsis Prediction Model** and stroke screening algorithms analyze EHR data in real time to prompt earlier workup and treatment, where minutes directly impact outcomes.

#### **4. Workload and Bed Management**

AI can predict **patient flow, boarding times, and surge periods**, helping charge nurses and administrators allocate staff and resources proactively rather than reactively.

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**Important Clinical Boundaries**

| Principle                                      | Why It Matters                                              |
|------------------------------------------------|-------------------------------------------------------------|
| AI supports, never replaces, clinical judgment | Final triage decisions must remain with licensed clinicians |
| Algorithmic bias is a real risk                | Models must be validated across diverse patient populations |
| Transparency is essential                      | Clinicians should understand <i>why</i> a flag was raised   |
| Over-reliance is a patient safety concern      | Staff must remain trained to triage independently           |

**Bottom Line**

AI serves as a **force multiplier for clinical teams** — reducing cognitive load, surfacing non-obvious risk, and enabling faster intervention. Its value is greatest when integrated into existing workflows with clear human oversight, not as a standalone decision-maker.

Screenshot: Simulated Configuration (Prompt Technique)

***"Provide a concise, fact-based explanation of how AI can reduce hospital readmission rates. Do not include opinions or speculation."***

6:26 PM

**AI and Hospital Readmission Reduction**

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**What the Evidence Shows**

Hospital readmissions within 30 days are a measurable patient safety and cost concern. AI contributes to reducing them through the following documented mechanisms:

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**1. Predictive Risk Scoring at Discharge**

AI models analyze **structured EHR data** — including diagnosis codes, medication lists, prior admissions, lab trends, and social determinants of health — to generate individualized readmission risk scores before a patient is discharged.

**Validated example:** The **LACE Index** has been augmented with machine learning to outperform traditional scoring in identifying high-risk patients across conditions including heart failure, COPD, and pneumonia.

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## 2. Targeted Post-Discharge Follow-Up

Risk scores generated by AI are used to **allocate care management resources** — such as nurse callbacks, home health visits, and early outpatient appointments — to patients statistically most likely to return to the ED.

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## 3. Medication Reconciliation Support

AI tools flag **medication discrepancies and high-risk drug regimens** at discharge, a documented driver of preventable readmissions, particularly in elderly and multi-morbidity populations.

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## 4. Remote Patient Monitoring Integration

AI processes continuous data from **wearables and home monitoring devices** — such as daily weight in heart failure patients or oxygen saturation in COPD patients — and alerts care teams to early deterioration signals before they escalate to readmission.

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### Documented Outcomes

| Study / Program                                        | Finding                                                                                                    |
|--------------------------------------------------------|------------------------------------------------------------------------------------------------------------|
| CMS Hospital Readmissions Reduction Program + AI tools | Hospitals using predictive analytics reported measurable readmission rate decreases in targeted conditions |
| Heart failure AI monitoring (multiple health systems)  | Early alert systems reduced 30-day readmissions by 20–30% in published trials                              |
| NLP on discharge summaries                             | Identified follow-up gaps linked to readmission risk not captured by structured data alone                 |

### Defined Limitations

- Model performance varies by patient population and institution
  - Requires integration with existing EHR infrastructure
  - Social determinants of health data remain incompletely captured in most models
  - Human clinical review is required before acting on AI-generated risk flags
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## Summary

AI reduces readmissions by identifying high-risk patients earlier, enabling targeted interventions, and supporting continuity of care through monitoring — all grounded in analysis of clinical data at a scale and speed not achievable manually.

***“Higher Temperature (More Creative): Provide a concise, fact-based explanation of how AI can reduce hospital readmission rates. Do not include opinions or speculation.”***

## AI and Hospital Readmission Reduction

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### The Problem

In the U.S., approximately **1 in 5 Medicare patients** is readmitted within 30 days of discharge — costing the healthcare system over **\$26 billion annually**. Many readmissions are preventable with better post-discharge monitoring and intervention.

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### How AI Reduces Readmissions

**1. Predictive Risk Scoring at Discharge** AI models analyze EHR data — including diagnosis, medications, prior admissions, lab trends, and social determinants — to generate a readmission risk score before a patient leaves the hospital. High-risk patients can then be flagged for enhanced follow-up.

**2. Remote Patient Monitoring** AI processes data from wearables and home devices — heart rate, oxygen saturation, weight, blood pressure — and detects early deterioration signals, triggering clinical outreach before a crisis requires readmission.

**3. Medication Adherence Monitoring** Natural language processing (NLP) tools analyze patient-reported symptoms and pharmacy refill data to identify non-adherence patterns linked to readmission risk in conditions like heart failure and COPD.

**4. Optimized Discharge Planning** AI identifies gaps in discharge instructions, follow-up scheduling, and specialist referrals — variables statistically associated with preventable returns to the ED.

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#### Documented Evidence

- The **LACE Index**, enhanced with machine learning, has demonstrated improved readmission prediction accuracy over traditional scoring alone.
  - Studies published in *Nature Medicine* show AI models outperforming standard clinical tools in 30-day readmission prediction for heart failure patients.
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#### Key Fact

AI does not independently reduce readmissions — it enables **earlier, more targeted human intervention** by surfacing risk that would otherwise go undetected until readmission occurs.

***“Lower Temperature (more focused): Provide a concise, fact-based explanation of how AI can reduce hospital readmission rates. Do not include opinions or speculation.”***

#### AI and Hospital Readmission Reduction

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##### What the Evidence Shows

Hospital readmissions within 30 days are a major clinical and financial burden. AI reduces readmission rates through the following validated mechanisms:

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##### 1. Predictive Risk Scoring at Discharge

Machine learning models — such as those built on **LACE index variables** (Length of stay, Acuity, Comorbidities, ED use) — analyze EHR data to identify patients at high risk of readmission before they leave the hospital. Studies published in *JAMA* and *The Lancet Digital Health* have demonstrated these models outperform traditional scoring tools in predictive accuracy.



2. Post-Discharge Monitoring

AI-powered remote monitoring tools track **vital signs, medication adherence, and symptom reports** from discharged patients via wearables or mobile apps. Anomalies trigger automated alerts to care teams, enabling early intervention before readmission becomes necessary.

3. Medication Management

Natural language processing (NLP) tools flag **medication discrepancies** at discharge — a leading cause of preventable readmission, particularly in heart failure and post-surgical patients.

4. Care Coordination Optimization

AI identifies patients who would benefit from **follow-up calls, home health visits, or outpatient appointments**, and helps schedule these interventions within clinically appropriate timeframes.

Documented Outcomes

| Intervention                             | Reported Readmission Reduction                         |
|------------------------------------------|--------------------------------------------------------|
| AI risk stratification + care navigation | Up to 20–25% reduction (CMS-reported pilots)           |
| Remote patient monitoring with AI alerts | 15–18% reduction in CHF patients                       |
| NLP-driven medication reconciliation     | Up to 30% reduction in medication-related readmissions |

Clinical Boundary

These tools function as **decision support systems**. Clinician review and action remain required for all flagged cases. Outcomes vary based on implementation quality and patient population.

Screenshot: Multi-turn Interaction or Image Prompt

*“List three ways AI can help reduce the workload of nurses in hospitals.”*

Here are three ways AI can help reduce the workload of nurses in hospitals:

1. **Automated patient monitoring** — AI-powered systems can continuously track patient vitals (heart rate, blood pressure, oxygen levels) and alert nurses only when intervention is needed. This frees nurses from routine check-ins and lets them focus on direct patient care.
2. **Administrative task automation** — A significant portion of a nurse's shift is spent on documentation, scheduling, and data entry. AI can auto-populate electronic health records (EHRs), transcribe physician notes, and streamline shift handovers, dramatically cutting paperwork time.
3. **Clinical decision support** — AI tools can analyze patient data in real time to flag early warning signs of deterioration (e.g., sepsis risk, fall risk), suggest care protocols, and prioritize task lists. This helps nurses make faster, more informed decisions without having to manually sift through records.

Together, these applications allow nurses to spend more time on what matters most — compassionate, hands-on patient care.

***“Can you give a real-world example or tool that is currently used for one of those?”***

Sure! Here's a real-world example for **clinical decision support**:

Searched the web

Searched the web

A great real-world example is **UpToDate Expert AI**, used widely in hospitals today.

It's a generative AI-powered clinical reasoning tool grounded in evidence-based medical content. It helps clinicians — including nurses — think through cases, clarify concepts, and answer clinical questions in real time, much like consulting a trusted colleague. [Wolters Kluwer](#)

Here's what makes it practical for nurses specifically:

- It integrates with major EHR systems like Epic and Cerner, provides drug interaction alerts, and offers personalized treatment suggestions — all accessible via a mobile app for point-of-care use. [DevOpsSchool.com](#)
- Nurses can ask questions and get concise, traceable answers with inline citations, with the option to refine or follow up — reducing the time spent hunting through medical literature mid-shift. [Wolters Kluwer](#)

- It's trusted by over 3 million clinicians globally and is backed by content from more than 7,600 clinical experts. [DevOpsSchool.com](https://www.devopschool.com)

Beyond UpToDate, research backs up the broader impact of these tools. One study found that an AI deterioration algorithm cut the time it took nurses to contact senior staff and order tests, while a discharge support system reduced 30-day hospital readmissions from 22.2% to 9.4%. [PubMed](https://pubmed.ncbi.nlm.nih.gov/)

So the real-world impact is measurable — faster decisions, fewer errors, and more time for nurses to focus on patients rather than paperwork or research.



***“Analyze this image and explain what information it provides in a healthcare context.”***

Looking at this image, I can see a skin condition displaying the following visible characteristics:

**What's visible:**

- Raised, red/pink plaques on the skin
- Silvery-white scaling over the affected areas

- Irregular, well-defined borders
- Appears on the torso/side of the body

**Likely condition:** This presentation is visually consistent with **psoriasis** — specifically plaque psoriasis, the most common form. Key indicators include the thick, scaly, raised patches with that characteristic reddish base and silvery scale overlay.

**Healthcare context:** In a clinical setting, this image could be useful for:

- **Documentation** of a patient's condition at a point in time to track progression or treatment response
- **Dermatology education** — it's a textbook-quality example of plaque morphology
- **Differential diagnosis reference**, as it helps distinguish psoriasis from conditions like eczema (which tends to be less defined) or ringworm (which is circular)
- **Treatment planning** — the extent and thickness of plaques visible here would inform whether topical, phototherapy, or systemic treatment is appropriate

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⚠️ **Important note:** I'm an AI and cannot provide a medical diagnosis. If you or someone else has a skin condition like this, please consult a **licensed dermatologist** for proper evaluation and treatment.

## Part B: Apply What You Learned

### Question 1

AI tools like Gemini Pro Vision could help radiologists in many ways. AI can help by analyzing medical images, summarizing key findings, and prioritizing scans that show potential abnormalities. This is especially important for radiologists because they are often overwhelmed with large volumes of imaging. Any delays in review can slow down diagnosis and treatment. The biggest benefit is speed. AI can process images quickly and help identify issues earlier. Though, accuracy is still critical because even small errors could lead to misdiagnosis. As noted in the Unit 4 readings, AI in healthcare should function as decision support, not replacement, since “human oversight remains essential for safe and effective clinical use” (Kumar et al., 2023). It seems to me the best approach is a collaboration between AI and doctors, where AI improves efficiency while clinicians maintain final judgment.

## Question 2

Gemini Pro could improve patient triage by helping prioritize patients. It could prioritize patients based on their initial symptoms, medical history, and urgency level during check-in. This could significantly reduce wait times in emergency departments. These departments delays and overcrowding often impact patient outcomes. The biggest benefit is speed and better prioritization, but there are important ethical concerns. These issues concern around fairness and the risk of incorrect triage decisions. As highlighted in healthcare AI research, systems must be carefully monitored to avoid bias and ensure equitable treatment across patients (World Health Organization, 2023). AI should assist in triage decisions, but human oversight is still necessary to ensure accuracy and patient safety.

## Question 3

Gemini Pro could be integrated into clinical documentation. Gemini could help by transcribing patient interactions, summarizing visits, and automatically filling in structured medical notes. This would save time for healthcare providers. This would allow providers to focus more on direct patient care instead of administrative tasks. One of the biggest benefits is reducing how time-consuming documentation can be. Though, a key challenge is ensuring the AI fully understands the clinical context and does not misinterpret important details. As noted in healthcare AI research, errors in generated medical information can impact care quality, so accuracy is critical (Zhang & Boulos, 2023). Because of this, AI-generated documentation should always be reviewed by a healthcare professional to ensure it is correct and safe.

## Question 4

Gemini Pro and Vision could improve medication adherence in many ways. They could help by providing reminders, tracking patient behavior, and offering check-ins to ensure medications are being taken correctly. Vision capabilities could also help by scanning prescriptions or recognizing pills to confirm proper usage. The biggest benefit is improved health outcomes. Patients who follow their treatment plans are less likely to experience complications or hospitalizations. However, with this there are important privacy concern. This is especially related to how patient data is collected, stored, and monitored. As highlighted in healthcare AI discussions, protecting sensitive patient information is critical when implementing these systems (Yaraghi, 2024). AI should assist and monitor patients with proper oversight rather than operate without control.

## Question 5

The finance industry faces many of the same challenges as Harmony Health. This includes high workloads, delays in processing information, and the need to analyze large amounts of data quickly while maintaining privacy. One similar issue is the time it takes to review and process financial documents. This includes documentation such as loan applications, transaction reports, and compliance records. The clinical documentation solution used in healthcare could be adapted to finance by using AI to automatically summarize, analyze, and organize financial data. This would allow faster and more efficient decision making. This would also improve speed and reduce manual effort. Though it also raises concerns around privacy and ensuring that sensitive financial data is handled securely.

In healthcare specifically, additional AI applications go beyond what Harmony Health is addressing. For example, AI is being used in drug discovery to identify new treatments faster. Another example is in personalized medicine where they tailor care based on patient data. AI is also improving medical imaging by detecting diseases earlier and supporting doctors in making more accurate diagnoses. These examples show that while AI can greatly improve efficiency and outcomes, transparency, privacy, and oversight remain critical to ensure safe and responsible use across both industries.

# References

Kumar, P., Chauhan, S., & Awasthi, L. K. (2023). Artificial intelligence in healthcare: Review, ethics, trust challenges, and future research directions. *Engineering Applications of Artificial Intelligence*, 120, 105894. <https://doi.org/10.1016/j.engappai.2023.105894>

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